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FairLens: A Counterfactual Framework for Algorithmic Bias Mitigation

Under IEEE 7003-2024

Emmanuel Amista

Third Year — Bachelor of Science in Computer Science

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Dr. Catherine R. Alimboyong



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INTRODUCTION

The rapid integration of machine learning into high-stakes sectors — healthcare, finance, criminal justice, and employment — has fundamentally altered how institutions make decisions. Algorithms now determine who receives loans, who is considered for job interviews, and who is flagged as a public safety risk. While these systems offer efficiency gains, they carry an underexamined danger: they can silently perpetuate and amplify the structural inequalities embedded in their training data. This is not a theoretical concern. It is a documented, measurable pattern in deployed systems (Mehrabi et al., 2021).

This case study introduces FairLens, a live, interactive Counterfactual Fairness Pipeline built to operationalize the IEEE 7003-2024 Standard for Algorithmic Bias Considerations. Rather than discussing fairness in abstract terms, FairLens provides a working prototype — backed by a real FastAPI and Pandas server — that demonstrates bias detection, measurement, and mitigation through a reproducible three-step engineering pipeline. The system is designed for CS 323 at North Eastern Mindanao State University as a demonstration that algorithmic fairness is not an optional feature but an engineering requirement.

The problem FairLens addresses is structural: developers routinely optimize for accuracy, precision, and recall while ignoring whether their models produce equitable outcomes across demographic groups. IEEE 7003-2024 was established specifically to



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close this professional gap. FairLens is the implementation tool that applies that standard in a real coding environment.

BACKGROUND OF THE STUDY

Machine learning has become embedded in consequential decisions across healthcare, finance, employment, and criminal justice. The problem is structural: models learn from historical data that reflects decades of social inequality. Bias is rarely the result of a programmer's explicit intent, it is inherited from flawed sampling, underrepresentation of minority groups, and historically prejudiced ground-truth labels (Mehrabi et al., 2021; Barocas et al., 2023).

From a professional standpoint, developers frequently optimize for performance benchmarks — accuracy, precision, recall — without measuring fairness across demographic groups. This gap between functional correctness and ethical responsibility is the central problem this study addresses (Ferrara, 2024). IEEE 7003-2024 was established to close that gap by making bias assessment a mandatory engineering step, not an afterthought.

FairLens directly addresses this by implementing a counterfactual fairness pipeline: a system that detects disparate outcomes across protected groups and enforces that decisions remain consistent when protected attributes are hypothetically changed. The backend is a real FastAPI server computing live Disparate Impact scores and



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applying counterfactual mitigation via Pandas — transforming fairness from an abstract ethical principle into a measurable, verifiable API response (Barocas et al., 2023).

Three documented real-world cases provide the empirical foundation for this study and are prominently featured in the FairLens web platform:

- Representation Bias in Facial Recognition: Buolamwini & Gebru (2018) found that 67% of images in widely-used facial-recognition benchmark datasets depict lighter-skinned subjects. Commercial systems showed error rates of 34.7% for darker-skinned women versus 0.8% for lighter-skinned men — in products sold by IBM, Microsoft, and Face++.
- Hiring Algorithm Bias: Amazon's AI recruiting tool, trained on 10 years of male-dominated résumés, learned to penalise résumés containing the word "women's" and downgraded graduates of all-women's colleges. The tool was internally scrapped in 2018 after reviews confirmed systematic discrimination against female applicants (Reuters, 2018; Ferrara, 2024).
- Criminal Justice AI (COMPAS): ProPublica's 2016 investigation found that the COMPAS recidivism algorithm flagged Black defendants as future criminals at nearly twice the rate of white defendants — a 45% versus 24% false-positive rate. Race was encoded implicitly through proxies such as zip code and prior arrest history (Mehrabi et al., 2021).



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STATEMENT OF THE PROBLEM

Algorithmic bias in machine learning systems is a systemic, measurable, and preventable injustice. Despite this, the dominant culture of software development prioritizes functional performance over ethical accountability. The following specific problems motivate the design of FairLens:

1. Protected demographic groups receive disproportionately unfavorable decisions from AI systems — even when protected attributes are not explicitly provided — because the models learn to infer them through statistical proxies such as zip codes, résumé keywords, and prior institutional records.
2. Most developers do not measure Disparate Impact (DI) before deploying their models. There is no standardized, accessible, executable pipeline that bridges the abstract requirements of IEEE 7003-2024 with working code.
3. Simply removing protected attributes from the input data is insufficient to achieve fairness. Models can still encode bias through correlated proxy variables, requiring counterfactual — not merely statistical — approaches to mitigation (Barocas et al., 2023).
4. Future IT professionals lack hands-on exposure to bias detection and mitigation tools. Academic instruction in CS 323 and equivalent courses must include executable, standards-aligned prototypes — not purely theoretical frameworks.



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PROPOSED SOLUTION

FairLens proposes four concrete, standards-grounded policy and engineering solutions:

1. Mandatory Model Cards

Every AI system deployed in a consequential decision context must be accompanied by a model card documenting: training data demographics, known performance disparities across protected groups, intended and prohibited use cases, and Disparate Impact scores measured before deployment. This requirement aligns with Google's Model Cards framework and is supported by Ferrara (2024). Feasibility: low cost, high accountability impact. Limitation: self-reported unless independently audited.

2. Pre-processing Sanitization (IEEE 7003-2024 §5.2)

Before model training begins, all datasets must pass through a bias detection pipeline — specifically ALF360's reweighing algorithm — to ensure balanced representation across protected groups. Teams must document every sanitization step and the resulting DI values. This implements the pre-processing requirement of IEEE 7003-2024 §5.2 (Pessach & Shmueli, 2022). Feasibility: open-source tooling is available. Limitation: reweighing may reduce overall accuracy marginally.

3. Counterfactual Fairness Audit (What-If Testing)



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Every model must pass a counterfactual fairness test before release: for each prediction, the system asks "Would this decision change if the applicant's protected attribute were different?" If the answer is yes for a significant proportion of cases, the model must be retrained. This enforces the causal definition of fairness in Barocas et al. (2023) and is the core mechanism demonstrated in the FairLens prototype. Feasibility: computationally inexpensive on structured data. Limitation: more complex for deep learning models.

4. ACM Code of Ethics Alignment

All recommendations are grounded in the ACM Code of Ethics: Principle 1.2 (Avoid Harm) — by mandating pre-deployment bias testing; Principle 1.4 (Be Fair and Not Discriminate) — by enforcing equal treatment across demographic groups; and Principle 2.5 (Give Comprehensive and Thorough Evaluations) — by requiring comprehensive DI reporting. Developers who deploy AI in high-stakes domains bear professional responsibility for its discriminatory outcomes, whether or not they intended them.



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SYSTEM ARCHITECTURE

FairLens is a full-stack web application composed of three layers: a React-based frontend, a FastAPI + Pandas backend, and an in-memory synthetic dataset. The system is complemented by the FairLens AI Coach, an AI-powered multilingual chatbot powered by the Groq API using the LLaMA 3.1-8b-instant model.

Frontend Layer

The user interface is built with React 18, transpiled in-browser via Babel Standalone, and styled with Tailwind CSS and Font Awesome 6. The UI contains six main sections accessible via a sticky navigation bar: Home, Background, Case Studies, Prototype, Solutions, and Conclusion. A dark/light theme toggle persists user preference via localStorage. The prototype section features an interactive loan dataset table, three action buttons, two SVG Disparate Impact gauge rings, a gradient DI bar with animated pins, a pipeline status panel, and post-mitigation female approval comparison cards.



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Backend Layer

The backend is a Python FastAPI server running on localhost:8000. It exposes three REST API endpoints: GET /api/dataset (returns the 10 synthetic loan applicants as JSON), POST /api/detect-bias (receives the dataset, computes biased predictions and Disparate Impact score, and returns approval rates), and POST /api/mitigate (applies counterfactual fairness reweighing, returns fair predictions and the corrected DI score). The data is processed with Pandas and held entirely in memory — no database is required.

Data Layer

The dataset contains 10 synthetic loan applicants with the following columns: Income (in thousands), Term (in months), Gender (M or F), and true_label (the ground-truth approval decision). The biased model computes a score as $(\text{Income} \times 0.8) - (\text{Term} \times 0.6) - 12$ if the applicant is Female, with a threshold of 28 for approval. The fair model uses the same formula without the gender penalty, with a threshold of 25. This design deliberately exposes a hidden -12 point gender penalty to make the bias concrete and measurable.



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AI Coach (Chatbot)

The FairLens AI Coach is a floating chat widget accessible via a pulsing FAB button. It is powered by the Groq API (<https://api.groq.com/openai/v1/chat/completions>) using the llama-3.1-8b-instant model with temperature 0.35, max_tokens 450, and top_p 0.9. The chatbot automatically detects the user's language — English, Filipino/Tagalog, or Bisaya/Cebuano — and responds in kind. It includes five pre-loaded suggestion chips, a structured system prompt with verified reference URLs, and scoped guardrails that restrict it to algorithmic bias and AI fairness topics only.



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SYSTEM ARCHITECTURE DIAGRAM

The following table illustrates the layered architecture of FairLens:

USER LAYER — React 18 + Babel (in-browser transpile) · Tailwind CSS · Font Awesome 6		
Navigation Bar Home Background Case Studies Prototype Solutions Conclusion Dark / Light Theme Toggle	Prototype Section Loan Dataset Table Detect Bias Button Apply CF Fairness Button Reset Button DI Gauge Rings	FairLens AI Coach Groq API · LLaMA 3.1-8b-instant Auto language detection EN / Filipino / Bisaya Suggestion chips
↑ HTTP REST API (JSON)		
BACKEND LAYER — FastAPI + Pandas (Python) · localhost:8000		
GET /api/dataset Returns 10 synthetic loan applicants (Income, Term, Gender, true_label)	POST /api/detect-bias Receives dataset JSON Computes biased_pred for each row Returns DI score, approval rates	POST /api/mitigate Applies CF fairness (removes gender penalty) Computes fair_pred Returns fair DI score, fair approval rates
DATA LAYER — Synthetic In-Memory Dataset · 10 Loan Applicants		
Columns: Income (k) Term (months) Gender (M / F) true_label (0 / 1)	Biased Model Formula: score = (Income × 0.8) – (Term × 0.6) – 12 if Female Threshold > 28 → Approve	Fair Model Formula: score = (Income × 0.8) – (Term × 0.6) [Gender-blind] Threshold > 25 → Approve

* Asterisk (*) in the dataset table denotes scores after the –12 pt gender penalty is applied. The Fair Model removes this penalty entirely.



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The following table shows the complete synthetic dataset used by FairLens and the effect of the biased model formula on female applicants:

#	Income	Term	Gender	True Label	Biased Prediction
1	85k	12mo	M	Approve	Approve (score=55.6)
2	90k	15mo	M	Approve	Approve (score=63.0)
3	45k	30mo	M	Deny	Deny (score=18.0)
4	50k	28mo	M	Deny	Deny (score=23.2)
5	88k	10mo	M	Approve	Approve (score=64.4)
6	75k	20mo	F	Approve	Deny (score=36.0 → 24.0*)
7	62k	22mo	F	Approve	Deny (score=36.4 → 24.4*)
8	80k	18mo	F	Approve	Deny (score=53.2 → 41.2*)
9	48k	29mo	F	Deny	Deny (score=20.9 → 8.9*)
10	52k	31mo	F	Deny	Deny (score=22.9 → 10.9*)

Disparate Impact Formula: $DI = \text{Approval Rate (Female)} \div \text{Approval Rate (Male)}$. $DI \leq 0.8$ = Discriminatory (IEEE 7003-2024 violation). $DI \geq 0.9$ = Fair (IEEE 7003-2024 compliant).



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KEY LITERATURES SUPPORTING FAIRLENS

The following review synthesizes the key literature and studies that underpin the design rationale of FairLens. Each source is examined in terms of its main focus, its relationship to the chosen IT-related social issue of algorithmic bias in modern software systems, its key findings or conclusions, and the gaps or limitations it presents that FairLens seeks to address.

1. Ferrara (2024) — Fairness and Bias in Artificial Intelligence: A Brief Survey of Sources, Impacts, and Mitigation Strategies

This study focuses on providing a comprehensive survey of how bias enters artificial intelligence systems, the societal impacts of these biases, and the current strategies used by developers to mitigate them.

This paper directly relates to the core issue of algorithmic bias by explaining that AI is not inherently neutral. It supports the FairLens project's claim that bias stems from flawed training data and human cognitive biases, reinforcing the need for proactive ethical engineering.

The study concludes that bias primarily originates from data representation issues and algorithmic design choices. It finds that mitigation cannot be a one-time fix; it requires continuous auditing, the use of automated fairness metrics, and strict model documentation.



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As a survey paper, it provides a broad overview of mitigation strategies but lacks a specific, step-by-step engineering pipeline. The FairLens project fills this gap by turning these broad strategies into a concrete, repeatable workflow aligned with IEEE 7003-2024.

2. Pessach & Shmueli (2022) — Algorithmic Fairness

This study focuses on the technical implementation of algorithmic fairness, specifically reviewing the mathematical metrics used to measure bias — such as the Disparate Impact ratio — and the methods used to fix it via pre-processing and in-processing techniques.

It provides the exact technical foundation needed for the FairLens pipeline. It validates the project's approach of using pre-processing bias detection and establishing a target fairness threshold.

The authors conclude that there is no "one-size-fits-all" fairness metric because different mathematical definitions of fairness often contradict each other. They also found that applying fairness constraints frequently results in a trade-off, where making a model fairer may slightly reduce its overall predictive accuracy.

The study notes that navigating the "accuracy-fairness trade-off" is highly dependent on the context of the software. This limitation is highly relevant to the FairLens project, meaning the framework must include human oversight to ensure the model remains usable after bias is mitigated.



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3. Barocas, Hardt, & Narayanan (2023) — Fairness and Machine Learning: Limitations and Opportunities

This foundational text explores the limitations of automated decision-making and introduces advanced concepts of causality and counterfactual fairness in machine learning.

This study provides the central mechanism for the FairLens project: counterfactual fairness. It explains how to test whether an AI's prediction changes when sensitive attributes, like race or gender, are hypothetically altered.

The authors find that simply "blinding" an algorithm to race or gender does not work, because AI can still infer those traits through statistical proxies. They conclude that causal reasoning and counterfactual testing are much more robust defenses against discrimination.

The text focuses heavily on the mathematics and theories of fairness but does not heavily address specific professional governance standards in a local IT setting. The FairLens project addresses this by tying these mathematical concepts directly to the IEEE 7003-2024 professional standard.



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4. IEEE Computer Society (2024) — IEEE Standard for Algorithmic Bias Considerations (IEEE Std 7003-2024)

This formal standard provides a framework for identifying, analyzing, and mitigating bias in algorithmic systems throughout the software development lifecycle. This is the most critical regulatory study for FairLens, as the project is designed to operationalize this specific standard. It moves the discussion from "ethics-by-suggestion" to "ethics-by-requirement".

The standard establishes that developers must be accountable for the datasets they choose and the potential harms those datasets may cause to marginalized groups. It mandates that fairness must be a measurable requirement, not an optional feature.

While the standard provides the "rules," it does not provide the "tools". It tells developers what to achieve but does not provide a pre-packaged code framework to achieve it. FairLens serves as the implementation tool that applies these rules in a real-world coding environment.



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5. Mehrabi et al. (2021) — A Survey on Bias and Fairness in Machine Learning

This study creates a detailed taxonomy of the different types of biases that infect machine learning, tracking exactly where they enter during the data collection and modeling phases.

It directly relates to the issue by identifying specific biases — such as representation bias and measurement bias — that cause AI to produce discriminatory outcomes against minority groups.

The study concludes that bias is a systemic lifecycle problem. It finds that bias enters at the data generation phase and the algorithm building phase, meaning mitigation must occur at the start of development.

While the study perfectly categorizes the types of bias, it serves primarily as a diagnostic survey. The relevant limitation for the FairLens project is that knowing the type of bias is not enough; computer science students and professionals need a prescriptive engineering pipeline to execute the actual fix.

6. Buolamwini & Gebru (2018) — Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification

This study audited commercial facial-recognition systems from IBM, Microsoft, and Face++ using the Pilot Parliaments Benchmark, a dataset balanced for skin type and gender.



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The study revealed stark intersectional disparities: error rates for darker-skinned women were up to 34.7%, compared to just 0.8% for lighter-skinned men. These findings provide the empirical case study for representation bias prominently featured in the FairLens prototype interface.

The study concludes that benchmark datasets used to train commercial vision systems are systematically skewed toward lighter-skinned subjects (67% of the total), leading to compounding inaccuracies for underrepresented groups.

The paper focuses on audit and measurement rather than engineering remediation. FairLens addresses this by implementing a live pipeline that not only measures DI but actively mitigates the underlying bias.



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ETHICAL ALIGNMENT WITH ACM CODE OF ETHICS

FairLens is grounded in the ACM Code of Ethics and Professional Conduct, the primary ethical framework governing computer science practice globally. Three specific principles guide every design decision in the system:

Principle 1.2 — Avoid Harm

ACM Principle 1.2 requires computing professionals to take care that any unintended harms are identified and corrected. In the context of algorithmic decision-making, harm is not limited to software bugs — it includes discriminatory predictions that deny opportunities to protected demographic groups. FairLens operationalizes this principle by making it impossible to deploy the system without first running the bias detection step. The mandatory Detect Bias phase computes the Disparate Impact score before any predictions are used, creating a technical enforcement of the "avoid harm" requirement rather than relying on developer awareness alone.



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Principle 1.4 — Be Fair and Take Action Not to Discriminate

ACM Principle 1.4 explicitly prohibits the use of computing technologies for discriminatory purposes and requires professionals to take active steps to prevent discrimination. The entire FairLens pipeline is a technical implementation of this principle. The Apply CF Fairness step removes the hidden -12 point gender penalty from the scoring function, directly enforcing equal treatment between male and female applicants. The use of counterfactual fairness — which asks "would this decision change if the applicant's gender were different?" — aligns with the causal interpretation of non-discrimination in Barocas et al. (2023).

Principle 2.5 — Give Comprehensive and Thorough Evaluations of Computer Systems

ACM Principle 2.5 requires that computing professionals only endorse systems that have been thoroughly evaluated. FairLens enforces this by requiring that the Disparate Impact score reach or exceed 0.9 — the IEEE 7003-2024 compliance threshold — before the system is considered to produce fair predictions. The pipeline makes the evaluation visible and interactive: users see the biased DI gauge, the gradient bar showing where the score falls relative to the 0.8 and 0.9 thresholds, and the before-and-after comparison of female approval rates. Fairness is not implied — it is measured and displayed.



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Integration of IEEE 7003-2024 and ACM Ethics

The combination of IEEE 7003-2024 and the ACM Code of Ethics provides FairLens with both regulatory standing and professional ethical grounding. IEEE 7003-2024 mandates what must be measured and documented; the ACM Code of Ethics mandates why. Together, they form the professional obligation that FairLens turns into executable code. For computer science students, this demonstrates that ethical requirements are not soft guidelines — they are professional standards with technical implementations.



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HOW TO USE THE FAIRLENS PROTOTYPE

The FairLens prototype is a live, interactive web application accessible at fairlens-app.netlify.app. The following step-by-step guide describes how to navigate and operate the full bias detection and mitigation pipeline.

Prerequisites

The frontend requires no installation — it runs in any modern browser. The full backend pipeline requires the FastAPI server (`main.py`) running on `localhost:8000`. If the backend is not running, the prototype will display an error: "Could not reach FastAPI backend. Make sure `main.py` is running on port 8000."

Step 1 — Load the Dataset

On page load, the React frontend automatically sends a GET request to `/api/dataset`. The response populates the Loan Applicant Dataset table with 10 synthetic applicants, each showing: row number, Income (k), Term (months), Gender badge (blue for M, pink for F), True Label (ground-truth decision), and two empty columns for Biased Pred and Fair Pred. The Biased Pred and Fair Pred columns display "—" until the pipeline is run.



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Step 2 — Detect Bias

Click the purple "1. Detect Bias" button. The frontend sends a POST request to `/api/detect-bias` with the full dataset as JSON. The backend computes the biased model score for each applicant — applying the -12 point gender penalty to all female applicants — and returns biased predictions, the Disparate Impact score, and gender-disaggregated approval rates.

The interface updates as follows: the Biased Pred column fills with color-coded badges (green checkmark for correct approval, yellow for false approval, red X for denial, gray for correct denial); the Biased DI gauge ring animates to the computed value; the DI gradient bar shows a white pin at the biased score position; the Pipeline Status box turns red and displays: "⚠ Bias Detected — DI = [score] (below 0.8). Female approval rate: X% vs male: Y%."

Step 3 — Apply Counterfactual Fairness

Click the green "2. Apply CF Fairness" button. The frontend sends a POST request to `/api/mitigate` with the full dataset. The backend removes the gender penalty from the scoring function entirely, recomputes predictions using the fair model, and returns fair predictions and the corrected Disparate Impact score.

The interface updates as follows: the Fair Pred column fills with corrected badges showing the decisions the gender-blind model produces; the Fair DI gauge ring animates



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to the improved value; the DI gradient bar shows a green pin at the fair score position; an arrow (→) appears between the two gauge rings; two comparison cards appear showing Biased Female Approval % versus Fair Female Approval %; the Pipeline Status box turns green and displays: "✓ Mitigation Applied — DI: [biased] → [fair]. Gender penalty removed. Decisions are counterfactually fair. IEEE 7003-2024 compliant."

Step 4 — Interpret Results

Compare the Biased Pred and Fair Pred columns row by row to identify which applicants received different decisions after mitigation. Female applicants who were denied under the biased model but approved under the fair model represent the direct impact of the hidden gender penalty. Use the DI score to assess compliance: $DI \geq 0.9$ confirms IEEE 7003-2024 compliance; DI between 0.8 and 0.9 indicates marginal fairness requiring further improvement; $DI \leq 0.8$ confirms a discriminatory model.

Step 5 — Use the FairLens AI Coach

Click the pulsing green balance-scale FAB button in the bottom-right corner to open the AI Coach chat window. The chatbot accepts questions in English, Filipino/Tagalog, or Bisaya/Cebuano and detects the user's language automatically. Use the five pre-loaded suggestion chips to ask common questions: "What is Disparate Impact?", "Ano ang counterfactual fairness?", "Unsa ang IEEE 7003-2024?",



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"How does the FairLens pipeline work?", and "Explain the COMPAS case". The AI Coach cites specific academic references in every response and will redirect off-topic questions back to algorithmic bias and fairness topics.

Step 6 — Reset

Click the gray "Reset" button to clear all predictions and return the pipeline to its initial state. The dataset table reverts to showing "—" in the Biased Pred and Fair Pred columns, both DI gauges return to zero, the DI pins disappear, and the Pipeline Status box resets to the idle instruction message. The dataset remains loaded from the API, so the next pipeline run can begin immediately.

CONCLUSION

Algorithmic bias is not an edge case, a bug, or an inevitability. It is a systems failure, and systems failures can be engineered out of existence. The FairLens pipeline demonstrates that bias detection and mitigation are technically feasible, computationally inexpensive, and completable before deployment via a real FastAPI backend. There is no legitimate engineering reason to ship a discriminatory model to production.



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The three case studies — Amazon's hiring AI, the COMPAS recidivism scoring system, and commercial facial recognition — share a common root cause: teams optimized for accuracy without measuring fairness. IEEE 7003-2024 closes this gap by mandating Disparate Impact measurement and mitigation before any consequential AI system is deployed. The FairLens prototype makes this standard executable and interactive.

The tools are available. The standards are published. The professional obligation is clear under ACM Principles 1.2, 1.4, and 2.5. The only remaining question is whether engineers and organizations will choose to use them — or repeat the same preventable failures at industrial scale. For computer science students and future IT professionals, FairLens offers a concrete, standards-aligned answer: algorithmic fairness is not an optional feature. It is an engineering requirement.

Ultimately, the code we write defines who receives opportunity and who does not. That is not a social observation — it is a professional responsibility.



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